

Open networks and composition

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1 Overview

An open Bayesian network is a network with an interface: some variables are inputs, provided by the environment, and some are outputs, offered to it. A network whose input variables carry no mechanism of their own is exactly a causal theory with free inputs ([Fong 2012](#)), and the interface is made precise as a structured cospan ([Baez and Courser 2020](#)), the same construction AlgebraicJulia uses for open dynamical and epidemiological systems ([Libkind et al. 2022](#)). Open networks compose by gluing outputs to inputs, and the semantics of the composite is the composite of the semantics: the kernel of A composed with B is the kernel of A composed with the kernel of B, and likewise for the tensor (Propositions 2 and 3 of the package specification). This vignette splits the reference habitat network into an abiotic and a biotic component, composes

them back in several ways, refines a mechanism by substitution, and checks the propositions numerically.

2 Setup

```
using CategoricalBayesianNetworks

m = reference_habitat_model()
ref = reference_habitat_bn()
variable_names(ref)
```

```
7-element Vector{Symbol}:
 :Climate
 :Irrigation
 :SoilMoisture
 :GrazingPressure
 :Vegetation
 :HabitatQuality
 :Occupancy
```

3 Splitting the reference network

The abiotic component generates SoilMoisture from Climate and Irrigation; the biotic component reads SoilMoisture and generates GrazingPressure, Vegetation, HabitatQuality and Occupancy. In the biotic network SoilMoisture has no mechanism (closed = false leaves it exogenous).

```
abiotic = bayesnet(:Climate ⇒ [:dry, :normal, :wet], :Irrigation ⇒ [:low, :high],
                  :SoilMoisture ⇒ [:low, :medium, :high];
                  mechanisms = [:SoilMoisture ⇒ (:Climate, :Irrigation)])
biotic = bayesnet(:SoilMoisture ⇒ [:low, :medium, :high], :GrazingPressure ⇒
[:low, :high],
                :Vegetation ⇒ [:sparse, :moderate, :dense], :HabitatQuality ⇒
[:poor, :good],
                :Occupancy ⇒ [:absent, :present];
                mechanisms = [:GrazingPressure ⇒ (),
                              :Vegetation ⇒ (:SoilMoisture, :GrazingPressure),
                              :HabitatQuality ⇒ :Vegetation,
                              :Occupancy ⇒ :HabitatQuality],
                closed = false)
variable_name.(Ref(biotic), exogenous(biotic))
```

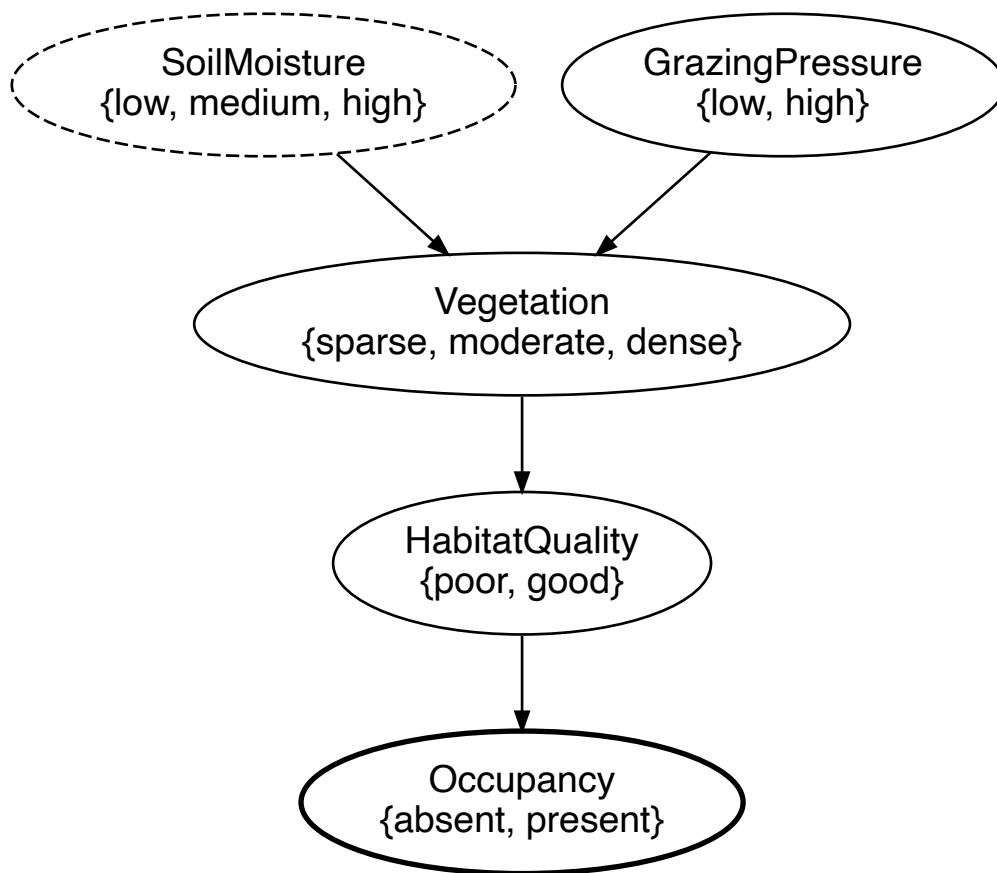
```
1-element Vector{Symbol}:
 :SoilMoisture
```

Open makes a network open by naming its inputs and outputs. The result is a Catlab structured cospan (Baez and Courser 2020): the apex is the network, the feet are VariableSpaces (the interface variables with their states) and the legs embed the feet into the apex.

```
A = Open(abiotic; outputs = [:SoilMoisture])
B = Open(biotic; inputs = [:SoilMoisture], outputs = [:Occupancy])
println(A)
println(B)
```

```
OpenBayesNet(3 variables, 3 mechanisms; Symbol[] -> [:SoilMoisture])
OpenBayesNet(5 variables, 4 mechanisms; [:SoilMoisture] -> [:Occupancy])
```

```
to_graphviz(B)
```



Inputs are drawn dashed and outputs bold. `input_space(B)` is the foot itself:

```
variable_names(input_space(B)), states(input_space(B), :SoilMoisture)
```

```
([:SoilMoisture], [:low, :medium, :high])
```

4 The typed-interface rule

Not every choice of interface is allowed. An open network must satisfy the typed-interface rule: input variables have no mechanism, the input leg is injective, every mechanism-free variable is an input, the apex is acyclic, and the legs preserve names, states and positions. Under this rule the pushouts behind composition never give a variable two mechanisms. Open checks the rule and rejects violations:

```
try
  Open(abiotic; inputs = [:SoilMoisture])      # SoilMoisture has a mechanism
catch e
  sprint(showerror, e)
end
```

"InterfaceError: rule 1 (every input-foot variable has no mechanism in the apex) is violated by variable(s) SoilMoisture (ids 3)"

```
try
  Open(biotic; outputs = [:Occupancy])         # SoilMoisture is exogenous but
  not an input
catch e
  sprint(showerror, e)
end
```

"InterfaceError: rule 3 (every mechanism-free apex variable is an input) is violated by variable(s) SoilMoisture (ids 1)"

5 Sequential composition and gluing

compose (or $\bullet$) is strict: the outputs of the first network must equal the inputs of the second, position by position, with the same states. The composite is the pushout of the two apexes along the shared interface.

```
C = compose(A, B)
inputs(C), outputs(C), sort(variable_names(apex(C)))
```

```
(Symbol[], [:Occupancy], [:Climate, :GrazingPressure, :HabitatQuality, :Irrigation,
:Occupancy, :SoilMoisture, :Vegetation])
```

The composite has no free inputs and one output, and its apex is the reference network again – gluing the two halves along SoilMoisture recovers exactly what was split:

```
canonicalize(apex(C)) == canonicalize(ref)
```

```
true
```

(ref is the structure alone, reference_habitat_bn(); the apex of C carries the NoRef references of its components, whereas syntax(m) carries the NamedRefs that bind_cpt assigned, so the comparison is made on the structure.)

A mismatch is reported with the position and the two sides:

```
B2 = Open(rename_variable(biotic, :SoilMoisture => :Moisture);
          inputs = [:Moisture], outputs = [:Occupancy])
try
  compose(A, B2)
catch e
  sprint(showerror, e)
end
```

```
"InterfaceMismatchError: compose: outputs(A) vs inputs(B): interfaces differ in
variable_name at position 1; left has :SoilMoisture, right has :Moisture"
```

glue composes along explicit pairs of names, which may differ (the glued variable keeps the name from the first network), and leaves the other interface variables in place:

```
G = glue(A, B2; along = [:SoilMoisture => :Moisture])
canonicalize(apex(G)) == canonicalize(ref)
```

```
true
```

6 Tensor and undirected wiring diagrams

otimes (or $\otimes$) places networks side by side; the apex is the disjoint union and the interfaces are concatenated. Names may repeat, which is fine for the structure but prevents evaluation until variables are renamed.

```
T = otimes(A, A)
inputs(T), outputs(T), nparts(apex(T), :Variable)
```

```
(Symbol[], [:SoilMoisture, :SoilMoisture], 6)
```

oapply composes along an undirected wiring diagram written with @relation: each box receives a network, its ports are the interface variables (inputs first), and shared junctions are glued. The typed-interface rule cannot be expressed in an undirected diagram, so it is checked afterwards.

```
uwd = @relation (occ,) begin
  abiotic(soil)
  biotic(soil, occ)
end
R = oapply(uwd, Dict(:abiotic => A, :biotic => B))
outputs(R), canonicalize(apex(R)) == canonicalize(ref)
```

```
([:Occupancy], true)
```

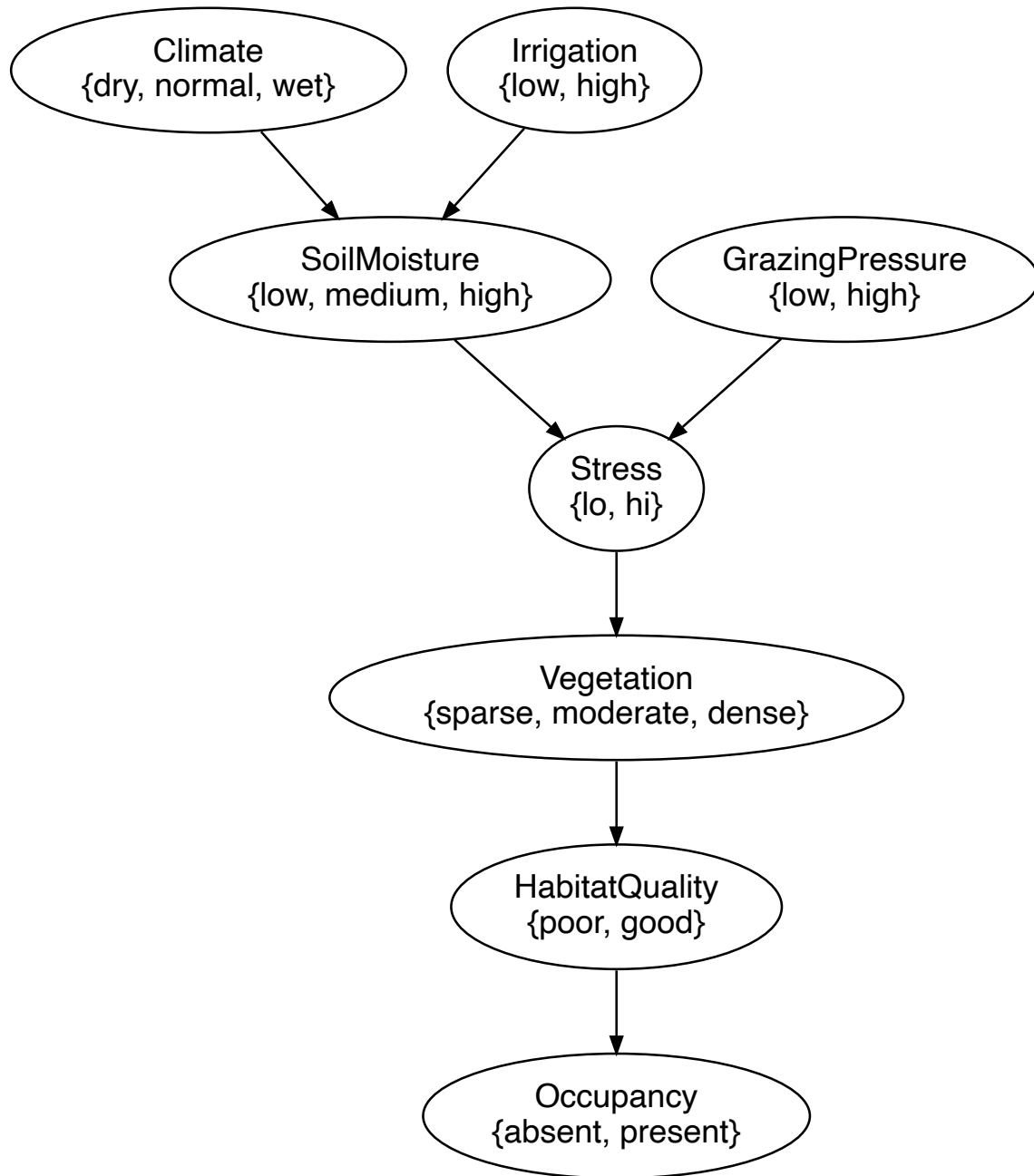
7 Substitution

substitute replaces a mechanism by an open network with the same interface: the parents as inputs, in order, and the target as output. Here the Vegetation mechanism is refined into a two-step pathway through a hidden Stress variable.

```
two_step = bayesnet(:SoilMoisture => [:low, :medium, :high], :GrazingPressure =>
[:low, :high],
                  :Stress => [:lo, :hi], :Vegetation => [:sparse, :moderate,
:dense];
mechanisms = [:Stress => (:SoilMoisture, :GrazingPressure),
              :Vegetation => :Stress], closed = false)
N = Open(two_step; inputs = [:SoilMoisture, :GrazingPressure], outputs =
[:Vegetation])
S = substitute(ref, :Vegetation_mechanism => N)
sort(variable_names(S)), mechanism_names(S)
```

```
([:Climate, :GrazingPressure, :HabitatQuality, :Irrigation, :Occupancy,
:SoilMoisture, :Stress, :Vegetation], [:SoilMoisture_mechanism,
:GrazingPressure_mechanism, :HabitatQuality_mechanism, :Occupancy_mechanism,
:Climate_mechanism, :Irrigation_mechanism, :Stress_mechanism,
:Vegetation_mechanism])
```

```
to_graphviz(S)
```



The same substitution can be done on the wiring diagram with Catlab's `substitute`, and reading the result back gives the same network:

```

d = to_wiring_diagram(ref)
b = findfirst(i -> box(d, i).value.name == :Vegetation_mechanism, box_ids(d))
dS = substitute(d, b, to_wiring_diagram(N))
canonicalize(first(from_wiring_diagram(dS))) == canonicalize(S)

```

true

On a `BayesModel` the substitution is recorded in the history, and kernels for the new mechanisms are bound afterwards.

8 Semantics of open networks: compositionality

`interpret` gives the kernel $\otimes$ inputs $\rightarrow \otimes$ outputs of an open network: the product of its mechanisms' kernels, summed over the hidden variables. Kernels are looked up by `KernelRef`, which survives composition because it is an attribute of the apex; the components above were built without references, so we look kernels up by target name instead.

```
by_name = Dict(x  $\Rightarrow$  kernel(m, x) for x in variable_names(ref))
kA = interpret(A, by_name)
kB = interpret(B, by_name)
kA.codom, kB.dom, kB.codom
```

```
(FiniteSpace(SoilMoisture{low,medium,high}),
FiniteSpace(SoilMoisture{low,medium,high}), FiniteSpace(Occupancy{absent,present}))
```

Sequential compositionality (Proposition 3 of the specification): the kernel of the composite is the composite of the kernels, so `interpret` is functorial.

```
interpret(compose(A, B), by_name)  $\approx$  compose(kA, kB)
```

```
true
```

Tensor compositionality (Proposition 2): the kernel of the tensor is the tensor of the kernels, so `interpret` is monoidal as well.

```
interpret(otimes(A, B), by_name)  $\approx$  otimes(kA, kB)
```

```
true
```

And the composite agrees with the closed model's own marginal, which is the reference evaluation of the previous vignette:

```
interpret(C, by_name)  $\approx$  marginal(m, :Occupancy)
```

```
true
```

9 Unrolling a dynamic network is gluing

The same machinery explains what `unroll` does to a `DynamicBayesNet`. A dynamic template is an initial network and a transition network related by the lag naming convention; `initial_slice` is the initial network renamed to slice 0 and `transition_slice(dbn, t)` the transition template seen from slice `t`, with the lagged variables exogenous. Gluing the outputs of one slice to the inputs of the next is exactly what `unroll` produces, up to renumbering of parts.

```
dbn = vegetation_herbivore_dbn()
A_0 = Open(initial_slice(dbn); outputs = [:Vegetation_0, :Herbivores_0])
B_1 = Open(transition_slice(dbn, 1); inputs = [:Vegetation_0, :Herbivores_0],
           outputs = [:Vegetation_1, :Herbivores_1])
G = glue(A_0, B_1;
         along = [:Vegetation_0 ⇒ :Vegetation_0, :Herbivores_0 ⇒ :Herbivores_0])
canonicalize(apex(G)) == canonicalize(unroll(dbn, 1))
```

true

`unroll` builds the network directly rather than by repeated pushouts, because that is faster and keeps the names predictable, but both test suites keep the two constructions in agreement. Dynamic networks are the subject of a vignette of `BayesianNetworks.jl`.

10 Summary

Open presents a network as a structured cospan, so composition is a pushout of ACSets rather than renaming by hand, and the typed-interface rule is the condition under which that pushout can never give one variable two mechanisms. `compose`, `glue`, `otimes`, `oapply` and `substitute` are all built from that one idea, and `interpret` sends each of them to the corresponding operation on finite stochastic kernels, which is what makes building a large model out of reviewed pieces safe; unrolling a dynamic network is one more instance of the same gluing. Serialisation and provenance, dynamic networks and model cards continue in `BayesianNetworks.jl`'s own vignettes.

11 References

Baez, John C., and Kenny Courser. 2020. “Structured Cospans.” *Theory and Applications of Categories* 35 (48): 1771–822.

Fong, Brendan. 2012. *Causal Theories: A Categorical Perspective on Bayesian Networks*. <https://arxiv.org/abs/1301.6201>.

Libkind, Sophie, Andrew Baas, Micah Halter, Evan Patterson, and James P. Fairbanks. 2022. "An Algebraic Framework for Structured Epidemic Modelling." *Philosophical Transactions of the Royal Society A* 380 (2233): 20210309. <https://doi.org/10.1098/rsta.2021.0309>.